



KIT-KALAI G NARKARUNANIDHI INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
(Accredited by NAAC&NBA with 'A' Grade)
Approved by AICTE & Affiliated to Anna University, Chennai)
Kannampalayam Post, Coimbatore-641402



Department of Computer Science and Business Systems

B19ADP301 – DATA SCIENCE LABORATORY

Name.....

Batch

Reg.No.....

Branch.....

Year



KIT-KALAI G N A R K A R U N A N I D H I I N S T I T U T E O F T E C H N O L O G Y
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Department of Computer Science and Business Systems

BONAFIDE CERTIFICATE

Name :

Roll No. : Reg. No.:

Branch : B. Tech – Computer Science and Business Systems

Certified that this is Bonafide record work done by Mr./Ms.....of

III – Year Computer Science and Business Systems during the academic year 2025-2026.

FACULTY IN-CHARGE

HOD

Submitted for the University Practical Examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

Instructions for Laboratory Classes

1. Enter the lab with record work book & necessary things.
2. Enter the lab without bags and footwear.
3. Footwear should be kept in the outside shoe rack neatly.
4. Maintain silence during the Lab hours.
5. Read and follow the work instructions inside the laboratory.
6. Handle the computer systems with care.
7. Shutdown the Computer properly and arrange chairs in order before leaving the lab.
8. The program should be written on the left side pages of the record workbook.
9. The record work book should be completed in all aspects and submitted in the next class itself.
10. Experiment number with date should be written at the top left-hand corner of the record work book page.
11. Strictly follow the uniform dress code for Laboratory classes.
12. Maintain punctuality for lab classes.
13. Avoid eatables inside and maintain the cleanliness of the lab.

VISION

To produce competent professionals to the dynamic needs of the emerging field of Computer Science and Business Systems.

MISSION

- To empower students with the knowledge and skills necessary to create intelligent systems and innovative solutions that address societal issues.
- Providing technical knowledge on par with Industry to the students through qualified faculty members having knowledge in recent trends and technologies.
- To produce competent engineers who are both professional and life-skills oriented.
- Providing opportunities for students to improve their research skills in order to address a variety of societal concerns through innovative projects.

PROGRAMME OUTCOMES (POs)

Students graduating from Computer Science and Business Systems should be able to:

PO1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of Computer Science and Business Systems problems.

PO2: Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and Computer Science and Business Systems.

PO3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations in the field of Computer Science and Business Systems.

PO4: Conduct Investigations of Complex Problems: Using research-based knowledge and Computer Science and Business Systems oriented research methodologies including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Computer Science and Business Systems Engineering activities with an understanding of the limitations.

PO6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and Sustainability: Understand the impact of the professional Computer Science and Business Systems Engineering solutions in societal and environmental contexts, and demonstrate the knowledge, and need for the sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance: Demonstrate knowledge and understanding of the Computer Science and Business Systems engineering and management principles and apply these to one's own work, as a member and leader in a team and, to manage projects in multidisciplinary environments.

PO12: Lifelong Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- **PEO1:** Graduates will perform well in their professional career by acquiring enough knowledge in mathematics, computer science and business systems to concord the industry engrossment.
- **PEO2:** Graduates will improve communication skills, business management skills, as well as educate on the service orientation principles for various business disciplines.
- **PEO3:** Graduates will demonstrate ethical and moral values, also involve in team work in their profession.

SYLLABUS

List of Experiments:

1. Working with Jupyter Notebook on fundamental Concepts.
2. Computations using NumPy functions – Computation on Arrays, Aggregation, Indexing and Sorting.
3. Data manipulations using Pandas – Handling of missing data and hierarchical indexing
4. Case study to demonstrate Curve Fitting.
5. Regression model for prediction
6. Classification Model
7. Find the outliers using plot.
8. Plot the histogram, bar chart and pie chart on sample data
9. Clustering model
10. Data Visualization using Matplotlib – Implementation of 2D plotting and 3D plotting

Total hours: 45

CONTENT

S. No	Title of the Experiment	Page No
1.	Working with Jupyter Notebook on fundamental Concepts. a) Creating and renaming a notebook b) Formatting the texts c) Creating a headers, ordered lists, bullet list and hyperlink d) Working with IPywidgets in the Jupiter Notebook (text and button widget, color piker and slider) e) Write a code to add two numbers using IPywidgets	
2.	Computations using NumPy functions a) Write a program to fin mean, standard deviation, variance and sum using Aggregate functions. b) Write a program to do sorting using functions. c) Write a program to indexing the array.	
3.	Data Manipulation using Pandas – Handling of Missing data a) creating a sample data frame and adding some missing values b) Drop rows or columns that have a missing value c) Drop based on a particular subset of columns d) Fill with a constant value e) Replace with the previous or next value	
4	Write codes in Python to Perform Curve Fitting.	
5	Regression model for data prediction a) Linear regression b) Polynomial regression c) Exponential regression	
6	Write a code to Solve A Simple Classification Problem with Python	
7	Write a code to find the outliers using plots	
8	Write a code to Plot following charts a) Histogram b) Bar chart c) Pie chart	

9.	Write to code to implement Clustering model	
10.	Data visualization using Matplotlib a) Implementation of 2D Plotting b) Implementation of 3D Plotting	

Content beyond syllabus

S. No	Title of the Experiment	Page No
1.	Plotting of Contour using contour() which only plots contour lines.	

S.No	Experiment	Prerequisites	Learning Objectives
1	For All Experiments	Basics of Python programming	1. To develop programs in Python using the role of constants, variables, identifiers, operators and other building blocks of python Language. 2. To create the python programs by using the conditional expressions and looping statements. 3. To develop the applications in python by using the concept of Array

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Ex.No.	1 a.	CREATING AND RENAMING A NOTEBOOK
Date		

AIM

To create and Rename Notebook in Jupyter Notebook.

ALGORITHM

- Step 1 : Open a Jupyter notebook using the Jupyter notebook command in command prompt.
- Step 2 : To create a Notebook in Jupyter, go to New and select Python 3.
- Step 3 : To rename the notebook, double click on the untitled at the top of the screen.
- Step 4 : In the pop up window enter a new notebook name that you want to add. Then click Rename.

Ex.No.	1 b.	FORMATTING THE TEXTS
Date		

AIM

To perform the formatting features of Jupyter in Jupyter Notebook.

ALGORITHM

- Step 1 : Open Jupyter Notebook
- Step 2 : To perform formatting features, first select Markdown Cell
- Step 3 : To make text bold, write text between the double underscores or double asterisks
- Step 4 : To make text italics write text between single underscore or single asterisk.
- Step 5 : Close the Jupyter Notebook.

Ex.No.	1 c.	CREATING HEADRERS, HYPERLINK AND LISTS
Date		

AIM

To create headers, Hyperlinks and lists in Jupyter Notebook

ALGORITHM

- Step 1 : Open Jupyter Notebook
- Step 2 : Select Markdown cell
- Step 3 : To make the text as a header, start the text using # symbol. The number of # symbols Depends upon the size of the header.
- Step 4 : To create a ordered list starts with 1. Use tab to make the suborder followed by the order.
- Step 5 : To create a bullet list starts with the dash(-) symbol, markdown cell converts dash into a Solid circle and asterisk (*) to a solid square.
- Step 6 : To attach a Hyperlink, place the name of link in square brackets[] and write link inside the parentheses().
- Step 7 : Close the Jupyter Notebook

Ex.No.	1 d.	WORKING WITH IPYWIDGETS
Date		

AIM

To write a code to working with common IPywidgets.

ALGORITHM

- Step 1 : Open Jupyter Notebook
- Step 2 : use the text widget to allow user to write the string
- Step 3 : use the button widget
- Step 4 : use color picker to select a color as per to your requirement
- Step 5 : use slider to find range and interval between two entities.
- Step 6 : close the Jupyter Notebook.

Ex.No.	1 e.	ADDITION OF TWO NUMBERS USING IPYWIDGETS
Date		

AIM

To write a code to perform the addition of two numbers using IPywidgets.

ALGORITHM

- Step 1 : Open Jupyter Notebook
- Step 2 : use IPywidgets to perform calculation
- Step 3 : using label get the input from user
- Step 4 : using display command display the input numbers
- Step 5 : using button perform addition
- Step 6 : display the output
- Step 7 : close the Jupyter Notebook

VIVA QUESTIONS

1. What does Jupyter stand for?
2. What is the basic element of a Jupyter Notebook?
3. What is special about Jupyter Notebook?
4. What are the three main types of Jupyter Notebook cell?
5. What is Jupyter Notebook file extension?

RESULT

Ex.No.	2 a.	AGGREGATE FUNCTIONS
Date		

AIM

To write a code to perform some arithmetic operations using Aggregate Functions.

ALGORITHM

- Step 1 : Start the program.
- Step 2 : Use NumPy to perform operations.
- Step 3 : Create a Array to do calculations like Mean, standard deviation, variance, sum and product of an array elements.
- Step 4 : use aggregate functions like np.mean (), np.sum (),np.prod(),np.var() and np.std() to perform operations.
- Step 5 : display the output..
- Step 6 : Stop the program.

Ex.No.	2 b.	SORTING FUNCTIONS
Date		

AIM

To write a code to perform sorting using functions.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Use NumPy and use any one of sorting function to perform sorting of elements
- Step 3 : Create an array to perform sorting
- Step4 : use numpy.sort() to perform sorting operation
- Step 5 : display the output
- Step 6 : Stop the program

Ex.No.	2 c.	INDEXING
Date		

AIM

To write a code to perform slicing and indexing.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Create an array to perform slicing and advanced indexing
- Step 3 : display the output of after slicing was done
- Step 4 : display the output of slicing using advanced index for column
- Step 5 : Stop the program

VIVA QUESTIONS

1. What is NumPy?
2. What are the uses of NumPy?
3. Why is NumPy preferred to other programming tools such as Idl, Matlab, Octave, Or Yorick?
4. What are the various features of NumPy?
5. List the advantages NumPy Arrays have over (nested) Python lists?

RESULT

Ex.No.	3 a.	HANDLING OF MISSING DATA
Date		

AIM

To write a code to create a sample data frame and some missing values

ALGORITHM

- Step 1 : Start the program
- Step 2 : create a data frame with 10 rows and 6 columns
- Step 3 : Add missing values that data frame, use loc method to select row and column
Combinations
- Step 4 : display the new data frame with missing values
- Step 5 : Stop the program

Ex.No.	3 b.	HANDLING OF MISSING DATA
Date		

AIM

To write a code to perform drop a rows and columns that having a missing value

ALGORITHM

- Step 1 : Start the program
- Step 2 : Display the new data frame with missing values
- Step 3 : To drop a row having any missing value then use dropna function
- Step 4 : To drop columns that have at least one missing value then use the axis parameter
- Step 5 : To drop a column or row that is full of missing values then use dropna function with how parameter
- Step 6 : To drop rows or columns with lots of missing values then use thresh parameter.
- Step 7 : Stop the program.

Ex.No.	3 c.	HANDLING OF MISSING DATA
Date		

AIM

To write a code to perform drop based on a particular subset of columns.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Display the data frame with missing values
- Step 3 : To drop a specific column based a subset use subset parameter of the dropna function
- Step 4 : Display the output after dropping a particular column
- Step 5 : Stop the program

Ex.No.	3 d.	HANDLING OF MISSING DATA
Date		

AIM

To write a code to fill the missing values with constant and aggregated value

ALGORITHM

- Step 1 : Start the program
- Step 2 : Display the data frame with missing values
- Step 3 : To fill the missing values use fillna function with one constant value
- Step 4 : To fill the missing value use fillna function with aggregated value such as mean, median or Mode
- Step 5 : Display the output
- Step 6 : Stop the program

Ex.No.	3 e.	HANDLING OF MISSING DATA
Date		

AIM

To write a code to replace the missing values with the previous/next value or another data frame

ALGORITHM

- Step 1 : Start the program
- Step 2 : Display the data frame with missing values
- Step 3 : To replace the missing values in a column with previous or next value in that column then use fillna function with method parameter
- Step 4 : To fill all missing values in data frame use another data frame by using fillna function
- Step 5 : Display the output
- Step 6 : Stop the program

VIVA QUESTIONS

1. Define the Pandas/Python pandas?
2. What is the important feature of panda?
3. Why a panda is used in Python?
4. How many types of data structures does pandas have?
5. What are the applications of pandas?

RESULT

Ex.No.	4	CURVE FITTING
Date		

AIM

To write a code to perform curve fitting

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import scipy optimization for using curve fit function
- Step 3 : Give the input to find coefficients to plot graph
- Step 4 : calculate sine and variance co-efficient
- Step 5 : plot the graph using Matplotlib
- Step 6 : Stop the program

VIVA QUESTIONS

1. What is curve fitting?
2. Why is curve fitting necessary?
3. What is difference between curve fitting and interpolation?
4. What module is useful for curve fitting in Python?
5. What are different types of curve fitting?

RESULT

Ex.No.	5 a.	LINEAR REGRESSION
Date		

AIM

To write a code to perform linear Regression.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import NumPy, random for NumPy and pyplot from Matplotlib (random from NumPy is used to create noise in a function)
- Step 3 : First create data set to do linear regression
- Step 4 : Estimate Beta coefficient
- Step 5 : Find predicted y values and R-Squared value
- Step 6 : Display the data
- Step 7 : Stop the program

Ex.No.	5 b.	POLYNOMIAL REGRESSION
Date		

AIM

To write a code to perform Polynomial Regression.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import NumPy, random for NumPy and pyplot from Matplotlib (random from NumPy is used to create noise in a function)
- Step 3 : First create data set to do linear regression
- Step 4 : Estimate Beta coefficient
- Step 5 : Find predicted y values and R-Squared value
- Step 6 : Display the data
- Step 7 : Stop the program

Ex.No.	5 c.	EXPONENTIAL REGRESSION
Date		

AIM

To write a code to perform Exponential Regression.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import NumPy, random for NumPy and pyplot from Matplotlib (random from NumPy is used to create noise in a function)
- Step 3 : First create data set to do linear regression
- Step 4 : Estimate Beta coefficient
- Step 5 : Find predicted y values and R-Squared value
- Step 6 : Display the data
- Step 7 : Stop the program

VIVA QUESTIONS

1. What is regression prediction model?
2. When would you use a regression model?
3. Which regression model is best?
4. What are two major advantages for using a regression?
5. What are some real life examples of regression?

RESULT

Ex.No.	6	CLASSIFICATION MODEL
Date		

AIM

To write a code to solve a Simple Classification Problem using k-NN-Classifier with Python

ALGORITHM

- Step 1 : Start the program
- Step 2 : Create a data set and plot graph
- Step 3 : Create a color maps
- Step 4 : Plot the decision boundary by assigning a color in the color map to each mesh point
- Step 5 : Put the result into a color plot
- Step 6 : Plot training points using k-NN classifier and display the output
- Step 7 : Stop the program

VIVA QUESTIONS

1. What is classification model in data science?
2. What types of Classification Algorithms do you know?
3. What is a confusion matrix?
4. What is the concept behind decision tree model ?
5. What is the concept behind random forest classification model?

RESULT

Ex.No.	7	FINDING OF OUTLIERS
Date		

AIM

To write a code to find outliers in the dataset

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import NumPy, pandas and Matplotlib to plot box plots
- Step 3 : Create a array with random integers between 1 to 20
- Step 4 : Find outliers
- Step 5 : Plot the graph
- Step 6 : Stop the Program

VIVA QUESTIONS

1. What do outliers mean?
2. What is a real life example of an outlier?
3. How do you find outliers in Python?
4. Why should we remove outliers?
5. Does removing outliers increase accuracy?

RESULT

Ex.No.	8 a.	HISTOGRAM
Date		

AIM

To write a code to plot histogram

ALGORITHM

- Step 1 : Start the program
- Step 2 : import NumPy and Matplotlib
- Step 3 : use any style to plot graph
- Step 4 : pass the mean, Sigma and size of the array as arguments in array
- Step 5 : using ax.hist () build the histogram.
- Step 6 : Display the output
- Step 7 : stop the program

Ex.No.	8 b.	BARChart
Date		

AIM

To write a code to Plot the bar chart.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Create a data to plot bar chart
- Step 3 : After that input the data as NumPy arrays
- Step 4 : Calculate the average and mean
- Step 5 : Create arrays for the plot
- Step 6 : Build the plot
- Step 7 : Display the plot
- Step 8 : Stop the program

Ex.No.	8 c.	PIECHART
Date		

AIM

To write a code to plot pie-chart

ALGORITHM

- Step 1 : Start the program
- Step 2 : Create a data to plot pie chart
- Step 3 : Use ax.pie () method plot pie chart
- Step 4 : Use ax.axis ('equal') to draw a pie chart as a circle
- Step 5 : Using labels and sizes specify values accordingly
- Step 6 : Use autopct to calculate percentage
- Step 7 : if you want to specify any pie pieces can be highlighted by exploding them out use explode
Display the output
- Step 9 : Stop the program

VIVA QUESTIONS

1. What is histogram in data science?
2. Why we use histogram in data science?
3. What is the advantage of histogram?
4. What are the applications of histogram?
5. What is the use of histogram?

RESULT

Ex.No.	9	CLUSTERING MODEL
Date		

AIM

To write a code to implement k-means algorithm

ALGORITHM

- Step 1 : Start the program
- Step 2 : Generate a dataset of groups in 2-dimensional space using sklearn
- Step 3 : Define a function that calculates Euclidean distances
- Step 4 : Initialize the centroids randomly and to iterations
- Step 5 : push current centroids to previous , reassign centroids as mean of the points
- Step 6 : Display the output
- Step 7 : Stop the program

VIVA QUESTIONS

1. What is a clustering model?
2. What is the importance of clustering in data science?
3. Is KNN and K-means same?
4. What is the application of clustering?
5. What are the clustering algorithms are available?

RESULT

Ex.No.	10 a.	IMPLEMENTATION OF 2D PLOTTING USING MATPLOTLIB
Date		

AIM

To write a code plot 2D Graphs.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import Matplotlib package
- Step 3 : Create an empty canvas
- Step 4 : Label the axes
- Step 5 : Create x, y as an array of numbers
- Step 6 : Create a 2D data scatter plot
- Step 7 : Show the 2D plot
- Step 8 : Stop the program

Ex.No.	10 b.	IMPLEMENTATION OF 3D PLOTTING USING MATPLOTLIB
Date		

AIM

To write a code plot 3D Graphs.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import Matplotlib package
- Step 3 : Create an empty canvas
- Step 4 : Label the axes
- Step 5 : use the gca function to plot 3D Projection
- Step 5 : Create a sine curve
- Step 6 : Show the 3D plot
- Step 7 : Stop the program

VIVA QUESTIONS

1. What is data visualization used for?
2. How many types of data visualization are there?
3. What do you mean by visualizing errors?
4. What process would you use to transform raw data into visualization?
5. What is depth cueing in visualization?

RESULT

Ex.No.	1.	CONTOUR PLOT USING MATPLOTLIB
Date		

AIM

To write a code plot contour using contour() method.

ALGORITHM

- Step 1 : Start the program
- Step 2 : Import Matplotlib package
- Step 3 : use contour method to plot diagram
- Step 4 : create 2D grid of features
- Step 5 : plot contour lines
- Step 5 : display the output
- Step 6 : Stop the program

PROGRAM

OUTPUT

VIVA QUESTIONS

1. What are the 3 main qualities of a contour line?
2. What is contour function?
3. How do you calculate a contour plot?
4. What is the use of `contourf()` function?
5. What are two applications of contour lines?

RESULT